VC Fundraising Pipeline Intelligence

Viva cheat sheet · 15 Oct 2026 · Live: vc-pipeline-intelligence.vercel.app · GitHub: sisonpiy-droid/vc-pipeline-intelligence

Say this first (60 seconds)

"Vikram is a first-time founder raising a Seed round for a B2B logistics SaaS. He has 50 VCs and 32 angels to chase and no time to watch the news for all of them. Every Monday and Thursday my system searches the public web for each investor's activity in the last 30 days. Gemini Flash reads the evidence and judges how relevant it is. **Fixed code, not the AI, turns that into a 0 to 100 follow-up score** and one recommended action. Only four columns are written back to Supabase, so his notes and contact history can never be touched. He opens the app and sees who to email first, why, with the source link and the exact quote."

Your 10 minutes

`1 min`	The pitch above
`3 min`	**Live demo:** Home (answer first) → click the top investor (score breakdown + quote + source) → Preview email (Calendly link) → Runs & audit (today's cron, credits, "failed is never quiet") → API reference (Integrations panel)
`1 min`	Components Map: follow one investor left to right
`5 min`	Questions (pages 4 and 5). If you are short of time, drop to the pitch plus the two sentences on the write boundary.

Five sentences to remember

1. **Evidence first:** no source link and verbatim quote, no recommendation.
2. **AI judges, code decides:** Gemini gives relevance 0 to 100; score and action are arithmetic in code.
3. **A refresh can only write four columns:** enforced in code, in the database role, and in tests.
4. **A failed search is never called "quiet":** unknown and quiet are different states.
5. **A run is data, not a request:** a durable queue survives serverless time limits and crashes.

Numbers

82
investors: 50 VC firms + 32 angels

30/30/20/15/5
recency / relevance / evidence / fit / relationship

75 · 55 · 40
reach out now ≥ 75 · prepare 55 to 74 · relevance floor 40

30 days
evidence window; undated pages dropped

4
columns a refresh may write

Mon + Thu 09:30 IST
cron 0 4 * * 1,4 UTC; may start ~1 h late (free plan)

3 items · 240 s
per claim · budget inside the 300 s function limit

900
search-credit budget a month; skip if checked < 36 h ago

15
API routes

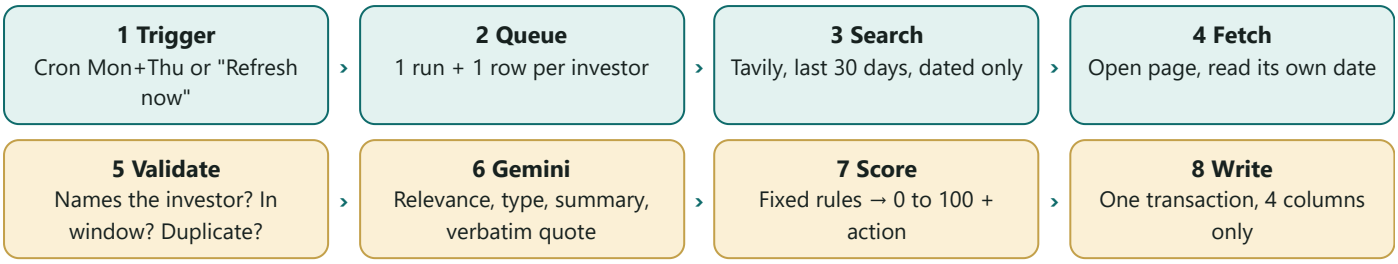
485 + 40
unit + browser tests, all passing

9
database migrations

95 vs 18
fresh relevant signal vs fresh irrelevant one (same investor)

Before you go in: open `/runs` and check that today's 09:30 IST cron ran (the 12 Oct run searched nothing because all 82 were checked within 36 h; today is the first run that really searches). Keep `/api/health` open in a tab.

The workflow



1. **Trigger.** Vercel Cron calls `/api/cron/refresh` with a secret. The planner skips investors checked in the last 36 hours and stops at the monthly credit budget.
2. **Queue.** A `runs` row plus one `run_items` row per investor. Each invocation claims 3 at a time (`FOR UPDATE SKIP LOCKED`) for 240 s, then hands the rest to a new invocation. A dead function is reclaimed after 5 minutes.
3. **Collect and validate.** One Tavily search per investor; every page is fetched safely and its date is read from the page itself. Rejected: wrong investor, outside 30 days, profile or directory pages, duplicates.
4. **Gemini Flash** returns structured JSON. Its quoted evidence must appear **word for word** in the fetched page, or the item is thrown away.
5. **Score and write.** Code computes the score; one database transaction as the restricted role `vc_refresh` saves the evidence and updates four columns. A failed item keeps its previous score and is shown as "couldn't check yet", never "quiet".
6. **Deliver.** The web app shows the ranked queue and weekly brief; a daily Telegram message (11:00 IST) summarises it; the founder sends mail from his own app. The app never sends email.

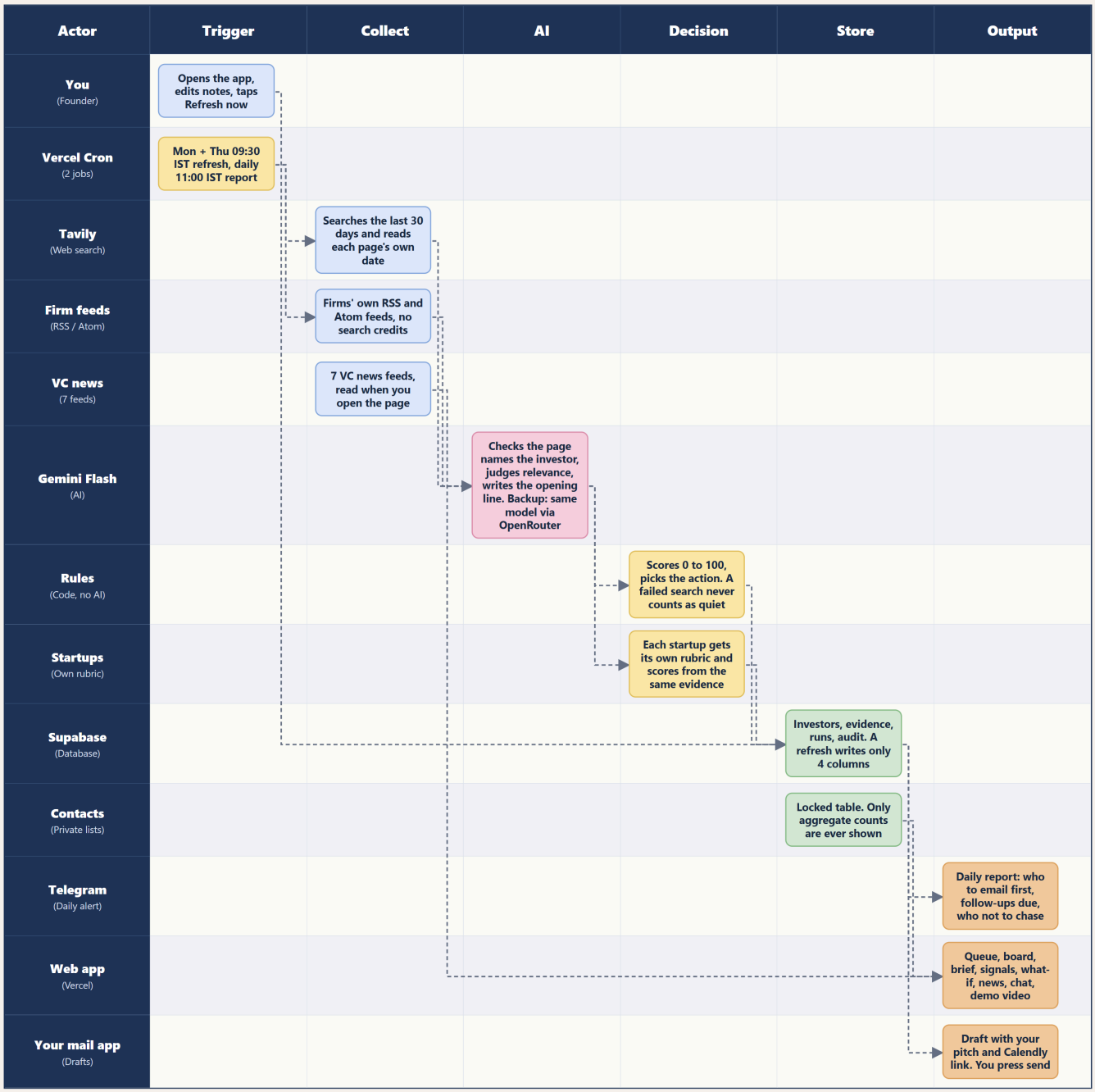
Why this workflow

CHOICE	REASON	REJECTED
Durable queue	82 investors × (search + fetches + AI) is minutes of work; a serverless function can be killed at 300 s. A run stored in the database is resumable and auditable.	One long request
AI judges, code scores	Models read text well but are poor at arithmetic and consistency. Fixed rules are explainable, testable and repeatable.	Model-written scores, dates or links
Verify the evidence	The quote must be in the page and the date must come from the page, so hallucinated sources cannot count.	Trusting the model
Database role for the write boundary	"Never touch notes" is the hardest requirement. A rule the database enforces cannot be broken by a coding mistake.	Careful code alone
Mon + Thu, free tiers	Even 3/4-day gaps; free Vercel cron; about 710 searches a month inside the free allowance. Vikram is bootstrapped.	Daily (costly), weekly (stale)
Plain pipeline, no agent	Steps are fixed, so a pipeline is cheaper, faster and fully auditable.	LangChain or an agent

Scope line: the core is refresh, score, write four columns safely, show the result. Chat, news, startups, cold search, Telegram and mail drafts were added on top; none writes to those four columns or the founder's data.

Components Map

Read left to right: Trigger → Collect → AI → Decision → Store → Output. Blue = inputs, pink = AI, yellow = rules, green = storage, orange = outputs.



APIs and services, and why

SERVICE	USED FOR	WHY THIS ONE
Tavily	Find recent pages per investor	Date filters; drops undated results; 1 credit per search; credit meter keeps it free
Gemini Flash	Judge evidence, opening line, chat	Required by the brief; fast, cheap, structured JSON output
Supabase Postgres	The database	Required; real roles, grants and row-level security
Vercel	Hosting, functions, cron	Push to main deploys; cron built in; free
RSS feeds	Firm feeds, news page	Free, dated, no key
Telegram Bot API	Daily report	Free, one HTTPS call
OpenRouter (backup)	Same Gemini Flash model if Google fails	Keeps the site alive when the free quota runs out; paid only then

Not used on purpose: Claude API at runtime (brief says Gemini), Firecrawl/Jina (a page's date and text are enough), Crunchbase, SerpAPI, MCP servers, any paid service. LinkedIn pages are never fetched.

My 15 routes: public (health, palette, chat) · passcode, off on the open demo (refresh now, save notes, startups) · secret, machines only (cron refresh, cron digest, continue, rescore). The secret never reaches a browser.

Backend

- **Next.js 16 route handlers** on Vercel, TypeScript, zod validation
- **Scoring:** recency 30 (full ≤ 3 days, then falls to 20% at day 30) + relevance 30 + evidence 20 + fit 15 + relationship 5. Gate: relevance ÷ 70; floor: 40. Irrelevant article: 72 → 18
- **Write boundary:** 4-field schema + role `vc_refresh` + before/after row tests (verified on the live DB)
- **Safe fetcher:** blocks internal addresses (SSRF), re-checks redirects, obeys robots.txt
- **Audit:** every run, source status, rejected candidate and reason is stored

Frontend

- **Server components** read Postgres directly: no database client or secret in the browser
- **Answer first:** home leads with who to email this week, each with its evidence
- **Honest states:** failed, partial and quiet look different
- **Tailwind 4, GSAP** motion, light/dark, phone widths tested, skip link, reduced motion respected
- **Pages:** queue, board, brief, pipeline, signals, what-if, news, startups, database, runs, API reference, video

Database: `investors` (profile · founder's notes · 4 signal columns) · `signals` · `runs / run_items` · `source_coverage` · `audit_logs` · startups tables · locked `investor_directory`. RLS on with no public policy, so Supabase's auto API exposes nothing. One connection string (`pg` directly, not supabase-js) so roles work and tests run on the real migrations (PGlite).

Most likely questions

1. How do you guarantee the founder's notes are never overwritten?
Three layers: a strict four-field schema in code, a database role that cannot read or write those columns, and tests that compare the whole row before and after. I verified the role on the live database.

2. Where is the AI used, and where not?
AI reads and judges text: involvement, event type, relevance, a summary, opening lines, chat. It never chooses dates, links or the score; those come from page metadata, the search provider and code.

3. How do you stop hallucination?
Dates come from the page, links from the search provider, output must be valid structured JSON, and the model's quote must appear word for word in the page I fetched, otherwise it is rejected.

4. How is the score worked out? Why 75 and 55?
Recency 30, relevance 30, evidence 20, fit 15, relationship 5; only relevance comes from Gemini. Relevance gates recency and evidence, and under 40 earns nothing. 75 needs a fresh, clearly relevant signal; an adjacent one lands in "prepare outreach". I chose them from a score grid; the first version used 70/50 and was too generous.

5. How do 82 investors fit in a 300-second serverless limit?
A run is rows in the database. Each invocation claims 3 investors for 240 s, then hands off to a fresh one. A dead function is reclaimed after 5 minutes; a lost hand-off is resumed by the next trigger. One active run is enforced by a unique index.

6. What does "quiet" mean, versus "failed"?
Quiet: every source answered and found nothing relevant in 30 days. Failed or partial: a search did not complete, so activity is unknown. A failed item writes nothing, keeps its old score, and the screen says "couldn't check yet".

7. How do you control cost? Why Monday and Thursday?
The planner skips investors checked in 36 hours, searches quiet ones about weekly, and stops at 900 credits a month. The free Vercel cron allows one a day with about an hour of jitter; Mon + Thu gives even 3/4-day gaps and about 710 searches a month.

HARD **8. The brief says Gemini only. Why is there an OpenRouter backup?**
It is the same Gemini Flash model through a different door, used only if Google fails (for example the free quota runs out). The product never switches model. It is optional, logged, and removing the key turns it off.

9. Is it secure? Anyone can open the site.
It is an open demo by choice so you can try it. Secrets stay on the server, machine routes fail closed without the secret, manual refresh has a cooldown, the chat is rate limited and cannot see notes or contacts. A real deployment would add sign-in.

10. You hold thousands of private contacts. Are they exposed?
No. They sit in a locked table (row-level security, no grants). The app shows counts and an organisation-level search only; a test fails if any name, email or phone ever appears in a result, and another that nothing in the app writes to that table.

11. Does the app send emails? Did you consider an agent or MCP?
It never sends: it builds a draft with the pitch and Calendly link and opens it in the founder's own mail app. I left out agents and MCP on purpose; a fixed pipeline is easier to test and audit, and an MCP server would be a second way into the data.

12. What is the weakest part?
Recall: one search per investor, no crawl of every firm's site. I am honest about it: coverage gaps are shown and "no signal" is only claimed when every source answered. Next: per-firm feeds or sitemaps, and recalibrating the weights from real outcomes.

HARD **13. What is not finished or not verified?**
The first scheduled run that actually searches is today at 09:30 IST; earlier runs prove the pipeline end to end (82/82, no errors). Telegram is connected and tested. The weights are reasoned from the brief, not yet calibrated on outcomes.

14. How would it scale to 5,000 investors?
The queue already splits work into small claims, so it scales sideways. Cost grows with searches and AI calls, so I would move to paid tiers, prioritise by score, and search quiet investors less often.